

ASSIGNMENT - 4.5

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Batch : 29

1. Suppose that you work for a company that receives hundreds of customer emails daily. Management wants to automatically classify emails into categories like "Billing", "Technical Support", "Feedback", and "Others" before assigning them to appropriate departments. Instead of training a new model, your task is to use prompt engineering techniques with an existing LLM to handle the classification.

Tasks to be completed are as below

a. Prepare Sample Data:

- Create or collect 10 short email samples, each belonging to one of the 4 categories.

b. Zero-shot Prompting:

- Design a prompt that asks the LLM to classify a single email without providing any examples.

- Example prompt:

“Classify the following email into one of the following categories: Billing, Technical Support, Feedback, Others. Email: ‘I have not received my invoice for last month.’”

c. One-shot Prompting:

- Add one labeled example before asking the model to classify a new email.

d. Few-shot Prompting:

- Use 3–5 labeled examples in your prompt before asking the model to classify a new email.

e. Evaluation:

- Run all three techniques on the same set of 5 test emails.
- Compare and document the accuracy and clarity of responses.

a. Sample Data (10 Emails)

No	Email Text	Category
1	I have not received my invoice for last month.	Billing
2	My payment was deducted twice from my account.	Billing
3	The app keeps crashing when I try to log in.	Technical Support
4	I cannot reset my password using the link provided.	Technical Support
5	Your new dashboard design looks amazing!	Feedback
6	I really appreciate the quick response from your team.	Feedback
7	Do you offer student discounts?	Others
8	What are your working hours during holidays?	Others
9	I was charged extra for a service I didn't use.	Billing
10	The software update failed to install properly.	Technical Support

b. Zero-Shot Prompting

Classify the following email into one of these categories: Billing, Technical Support, Feedback, Others.

Email: "The app keeps crashing when I try to log in."

c. One-Shot Prompting

Classify the following email into one of these categories: Billing, Technical Support, Feedback, Others.

Example:

Email: "I was charged twice for my subscription."

Category: Billing

Now classify:

Email: "I cannot reset my password using the link provided."

Category:

d. Few-Shot Prompting

Classify the following email into one of these categories:

Billing, Technical Support, Feedback, Others.

Example 1:

Email: "I have not received my invoice."

Category: Billing

Example 2:

Email: "The app crashes every time I open it."

Category: Technical Support

Example 3:

Email: "Your service is excellent!"

Category: Feedback

Example 4:

Email: "Do you have weekend support hours?"

Category: Others

Now classify:

Email: "My payment was deducted twice from my account."

Category:

e. Evaluation

Test Emails (Same 5 Used for All Methods)

1. I was charged extra for a service I didn't use.

2. The software update failed to install properly.
3. Your team did a fantastic job!
4. What are your business hours?
5. I cannot access my account after payment.

Observation:

1. **Zero-shot works well for clear cases but may confuse overlapping scenarios.**
2. **One-shot improves formatting consistency and classification accuracy.**
3. **Few-shot provides the most reliable and stable output.**
4. **More examples reduce ambiguity.**
5. **Prompt clarity directly affects response quality.**

Task2:

Travel Query Classification

Scenario:

A travel assistant must classify queries into Flight Booking, Hotel Booking, Cancellation, or General Travel Info.

Tasks:

- a. Prepare labeled travel queries.

- b. Apply Zero-shot prompting.
- c. Apply One-shot prompting.
- d. Apply Few-shot prompting.
- e. Compare response consistency.

a. Labeled Travel Queries (Sample Data)

No	Query	Category
1	I want to book a flight from Delhi to Mumbai next Monday.	Flight Booking
2	Can you reserve a hotel room in Goa for 3 nights?	Hotel Booking
3	I need to cancel my flight ticket for tomorrow.	Cancellation
4	What is the baggage allowance for international flights?	General Travel Info
5	Book a return ticket to Hyderabad for this weekend.	Flight Booking
6	Please cancel my hotel reservation for 15th June.	Cancellation
7	Find me a 4-star hotel near the airport.	Hotel Booking
8	What documents are required for domestic travel?	General Travel Info

b. Zero-Shot Prompting

Classify the following query into one of these categories:
Flight Booking, Hotel Booking, Cancellation, General Travel Info.

Query: "I need to cancel my flight ticket for tomorrow."

c. One-Shot Prompting

Classify the following query into one of these categories:
Flight Booking, Hotel Booking, Cancellation, General Travel Info.

Example:

Query: "Book a ticket from Chennai to Bangalore."

Category: Flight Booking

Now classify:

Query: "Can you reserve a hotel room in Goa for 3 nights?"

Category:

d. Few-Shot Prompting

Classify the following query into one of these categories:
Flight Booking, Hotel Booking, Cancellation, General Travel Info.

Example 1:

Query: "Book a ticket from Delhi to Pune."

Category: Flight Booking

Example 2:

Query: "Cancel my hotel reservation for tomorrow."

Category: Cancellation

Example 3:

Query: "Find a hotel near the beach in Goa."

Category: Hotel Booking

Example 4:

Query: "What is the check-in time for most hotels?"

Category: General Travel Info

Now classify:

Query: "I need to cancel my flight ticket."

Category:

e. Response Consistency Comparison

1. Cancel my ticket for next Friday.
2. Book a hotel room in Jaipur.
3. What is the refund policy for cancelled tickets?
4. I need a flight to Kolkata tomorrow.

Observation:

Zero-shot → Works for clear requests but may confuse refund-related queries.

One-shot → Improves structure and reduces ambiguity.

Few-shot → Most consistent and accurate.

More examples → Better clarity and stable output format.

3. Programming Question Type Identification

Scenario:

A coding help chatbot must classify queries into Syntax Error, Logic Error, Optimization, or Conceptual Question.

Tasks:

- a. Prepare coding-related user queries.
- b. Perform Zero-shot classification.
- c. Perform One-shot classification.
- d. Perform Few-shot classification.
- e. Analyze improvements in technical accuracy.

a. Sample Coding-Related Queries (Labeled Data)

No	Query	Correct Category
1	Why am I getting an unexpected indent error in Python?	Syntax Error
2	My program runs but gives incorrect output.	Logic Error
3	How can I reduce the time complexity of this algorithm?	Optimization
4	What is the difference between list and tuple in Python?	Conceptual Question
5	I forgot a semicolon and now it shows an error in C.	Syntax Error
6	My function always returns 0 even when input changes.	Logic Error
7	How can I make this code execute faster?	Optimization
8	What is recursion in programming?	Conceptual Question

b. Zero-Shot Classification

Classify the following query into one of these categories:
Syntax Error, Logic Error, Optimization, Conceptual Question.

Query: "My program runs but gives incorrect output."

c. One-Shot Classification

Classify the following query into one of these categories:
Syntax Error, Logic Error, Optimization, Conceptual Question.

Example:

Query: "I get a missing colon error in Python."

Category: Syntax Error

Now classify:

Query: "How can I improve the performance of this code?"

Category:

d. Few-Shot Classification

Classify the following query into one of these categories:
Syntax Error, Logic Error, Optimization, Conceptual Question.

Example 1:

Query: "I get an unexpected indent error."

Category: Syntax Error

Example 2:

Query: "My program compiles but produces wrong results."

Category: Logic Error

Example 3:

Query: "How can I reduce memory usage in this program?"

Category: Optimization

Example 4:

Query: "What is polymorphism in OOP?"

Category: Conceptual Question

Now classify:

Query: "My code works correctly but takes too long to execute."

Category:

e. Analysis of Improvements in Technical Accuracy

I get a NameError in Python.

My function returns wrong output for some inputs.

How do I make this algorithm faster?

What is abstraction in OOP?

Observation:

Zero-shot → Good for obvious keywords but weaker for subtle distinctions.

One-shot → Improves response format and consistency.

Few-shot → Highest technical accuracy and clearer category separation.

Examples significantly reduce ambiguity between Logic Error and Optimization.

4. Social Media Post Categorization

Scenario:

A social media analytics tool must classify posts into Promotion, Complaint, Appreciation, or Inquiry.

Tasks:

1. Prepare sample social media posts.
2. Use Zero-shot prompting.
3. Use One-shot prompting.
4. Use Few-shot prompting.
5. Analyze informal language handling.

1. Sample Social Media Posts (Labeled)

No	Post	Correct Category
1	Get 50% off on all products this weekend! Don't miss out!	Promotion
2	My order arrived damaged and no one is responding.	Complaint
3	Absolutely love the new update! Great job team!	Appreciation
4	Can someone tell me how to track my shipment?	Inquiry
5	Huge discounts available now, shop today!	Promotion

- | | | |
|---|---|--------------|
| 6 | Very disappointed with the late delivery. | Complaint |
| 7 | Thanks for the quick support, really helpful! | Appreciation |
| 8 | Is this product available in blue color? | Inquiry |

2. Zero-Shot Prompting

Classify the following social media post into one of these categories:
Promotion, Complaint, Appreciation, Inquiry.

Post: "My order arrived damaged and no one is responding."

3. One-Shot Prompting

Classify the following social media post into one of these categories:
Promotion, Complaint, Appreciation, Inquiry.

Example:

Post: "Huge sale live now! Grab your deals."

Category: Promotion

Now classify:

Post: "Can someone tell me how to track my shipment?"

Category:

4. Few-Shot Prompting

Classify the following social media post into one of these categories:
Promotion, Complaint, Appreciation, Inquiry.

Example 1:

Post: "Flat 40% discount today only!"

Category: Promotion

Example 2:

Post: "This is the worst service ever."

Category: Complaint

Example 3:

Post: "Thank you for the amazing support!"

Category: Appreciation

Example 4:

Post: "How do I change my account password?"

Category: Inquiry

Now classify:

Post: "Wow... waited 10 days for delivery "

Category:

5. Analysis of Informal Language Handling

"Loved itttt 🥰🔥 "

"Bruh this app crashes every time "

"Any idea when restock happening??"

"Flash sale!!! Hurry up guys!!"

Observation:

Improves consistency

Handles informal language better

Detects sarcasm and emotional tone

Provides highest classification reliability